GEMINIVIRUS DELIVERY OF CRISPR/CAS9 FOR PLANT GENOME ENGINEERING

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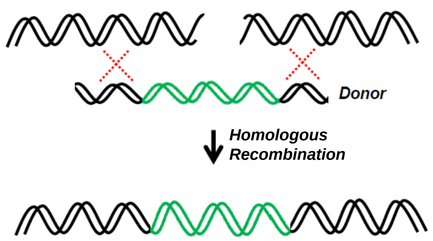
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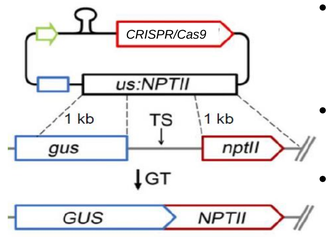
Abstract

With the characterization and optimization of genome editing reagents, like CRISPR/Cas9 (clustered, regularly interspaced, short palindromic repeats/CRISPR associated protein 9), the limiting factor for genome editing is the delivery for these reagents. Homologous recombination-mediated genome edits require a donor template of DNA for repair along with a protein containing a DNA-targeting domain and a nuclease domain (plus a guide-RNA for CRISPRs). Geminiviruses present a possible means of delivering these in high quantities. To further explore the potential of geminiviruses, the virus was used to deliver the reagents to repair a knocked out beta-glucuronidases (GUS) gene in *N. benthamiana*. Geminiviruses were constructed containing the necessary reagents, along with different guide-RNAs. Using *Agrobacterium* the geminivirus was infiltrated into *N. benthamiana* leaves and delivered on transfer-DNA to leaves. The effectiveness of each guide-RNA was determined by GUS staining. The pending results will allow further experimentation, optimizing geminiviruses as means of delivery of CRISPR/Cas9.

Introduction

- Genome edits include removing and inserting genes
- Done by creating double strand breaks in DNA
- Homologous recombination repairs DNA breaks
- CRISPR is a protein that, guided by a gRNA finds target DNA sequences and binds the double helix
- Cas9 cleaves DNA, creating a break





- Delivery of these proteins is a limiting factor
- Geminiviruses can deliver all needed reagents
  - genes, proteins and gRNAs
- The virus can replicate within the cell
  - Creating a high copy number of reagents
    - This experiment used transgenic tobacco containing a broken *gus:nptII* gene
    - The gene was to be repaired using the CRISPR/Cas9 system
    - The product of the gene can be stained blue

The purpose of this experiment was to demonstrate genome modification using CRISPR/Cas9 delivered via geminiviruses

- Four different gRNAs were tested to identify active gRNA
  - High gene targeting activity
  - Low variability

Once effective gRNAs are identified and the CRISPR/Cas9 system has been demonstrated in the geminivirus, different structures of the virus and ways of expressing CRISPR/Cas9 can be tested.

Results

To determine the effectiveness of a gRNA, a good control is needed.
So what makes a good control?

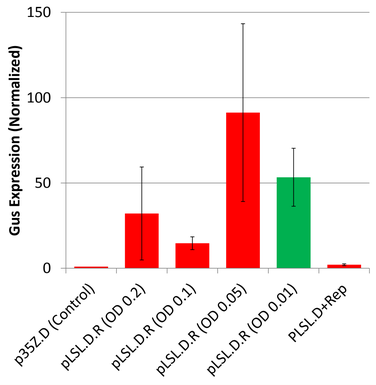
Similarity to Treatment

pLSLZ.D.R
Most similar plasmid: a zinc finger nuclease (ZFN) confirmed to be functional (Baltes et al., 2014), donor DNA, and replication protein on a geminivirus vector. Four concentrations of *Agrobacterium* infiltrated.

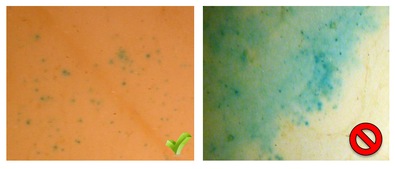
pLSL.D+pREP
Lower Expression: Donor molecule on geminivirus vector with replication protein expressed transiently.

p35Z.D
ZFN and donor DNA expressed on a promoter. A positive control.

High Activity & Low Variability



Distinguishable Spots



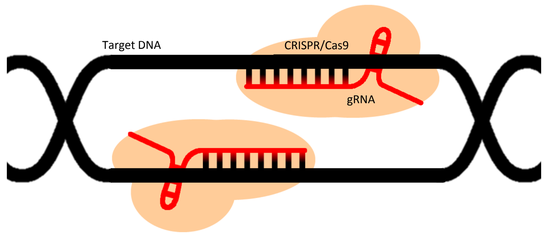
Tentative Results



GUS expression has been observed for all three of the gRNAs. The leaves still need to be further de-stained and then analyzed for activity and variability

Discussion

- Genetic modification in plants using a CRISPR/Cas9 was demonstrated
- A good positive control was identified
- Three gRNAs appear active
- With functional gRNAs more sophisticated experiments can be performed to improve the CRISPR/Cas9 system
- For the CRISPR/Cas9 has a notable drawback:
  - It can bind to the wrong places and cut there
- These off-target effects can create mutations
- One method to reduce off-target effects:
  - Express CRISPR/Cas9 transiently
- Another method:
  - A CRISPR/Cas9 nickase system

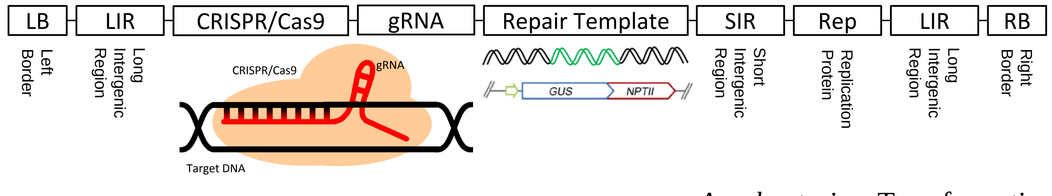


The CRISPR/Cas9 nickase system (seen above)

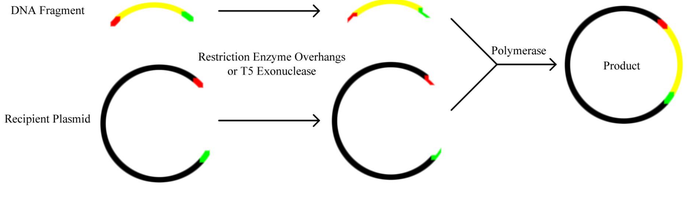
- Uses a mutant Cas9
  - Only cuts one side of the double helix
- Using two gRNAs, two nicks are made at the double helix, creating a double strand break
- If one CRISPR/Cas9 nicks the wrong place there is no break, no deleterious mutations
- Now can demonstrating this with geminiviruses

Methods

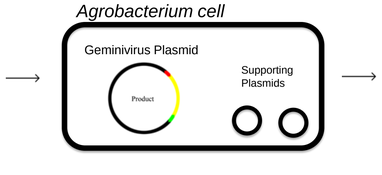
Plasmid Construction



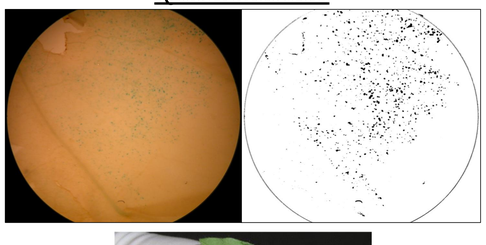
Gibson Assembly & Golden Gate Assembly



Agrobacterium Transformation



Quantification



Leaf Infiltration



Acknowledgements



References

Baltes, N. J., Gil-Humanes, J., Cermak, T., Atkins, P. A., & Voytas, D. F. (2014). DNA replicons for plant genome engineering. *The Plant Cell*, *26*(1), 151–63. doi:10.1105/tpc.113.119792

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